Condition Numbers & Matrix Norms — Concise Reference

Based on standard results in *Higham (2002)*, *Golub & Van Loan (2013)*, *Trefethen & Bau (1997)*.

A. Definitions (Norms & Condition Numbers)

Definition 1 (Matrix norms)**.** *Let* $A \in \mathbb{R}^{m\times n}$*. The* spectral norm *is* $\|A\|_2 = \sigma_{\max}(A)$*; the* Frobenius norm *is* $\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2\right)^{1/2} = \left(\sum_k \sigma_k^2\right)^{1/2}$*; the* induced 1- *and* ∞*-norms are* $\|A\|_1 = \max_j \sum_i |a_{ij}|$ *and* $\|A\|_\infty = \max_i \sum_j |a_{ij}|$*.*

Definition 2 (Condition number for solving $Ax = b$)**.** *For a nonsingular* $A \in \mathbb{R}^{n\times n}$ *and a given matrix norm* $\|\cdot\|$*,*

$$\kappa_{\|\cdot\|}(A) \;=\; \|A\|\,\|A^{-1}\|.$$

Common cases:

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)}, \qquad \kappa_F(A) = \|A\|_F \left\|A^{-1}\right\|_F, \qquad \kappa_1(A) = \|A\|_1 \left\|A^{-1}\right\|_1, \quad \kappa_\infty(A) = \|A\|_\infty \left\|A^{-1}\right\|_\infty.$$

Remark 1. $\kappa_{\|\cdot\|}(A)$ measures worst-case *forward-error amplification* under small data perturbations for the problem $Ax = b$ using the chosen norm.

B. Relationships and Inequalities Among Norms and κ

Norm comparisons (for $A \in \mathbb{R}^{m\times n}$):

$$\|A\|_2 \;\le\; \|A\|_F \;\le\; \sqrt{\operatorname{rank}(A)}\,\|A\|_2 \;\le\; \sqrt{\min\{m,n\}}\,\|A\|_2\,.$$

$$\|A\|_2 \;\le\; \sqrt{\|A\|_1\,\|A\|_\infty}, \qquad \|A\|_F \;\le\; \sqrt{\|A\|_1\,\|A\|_\infty}.$$

Condition-number comparisons (square $n \times n$ nonsingular):

$$\kappa_2(A) \;\le\; \kappa_F(A) \;\le\; n\,\kappa_2(A), \qquad \sqrt{\kappa_1(A)\,\kappa_\infty(A)} \;\ge\; \kappa_2(A).$$

C. Core Conditioning/Error Theorems (Square Real Matrices)

Theorem 1 (First-order forward error bound)**.** *For $Ax = b$ with perturbations δA and δb such that $A + \delta A$ is nonsingular and terms of order $\mathcal{O}(\|\delta \cdot\|^2)$ neglected,*

$$\frac{\|\delta x\|}{\|x\|} \lesssim \kappa_{\|\cdot\|}(A) \left(\frac{\|\delta A\|}{\|A\|} + \frac{\|\delta b\|}{\|b\|} \right).$$

Theorem 2 (Backward error $\Rightarrow$ forward error)**.** *If a computed $\widehat{x}$ satisfies $(A+\Delta A)\widehat{x} = b$ with $\|\Delta A\| \le \eta \|A\|$, then*

$$\frac{\|\widehat{x} - x\|}{\|x\|} \le \frac{\kappa_{\|\cdot\|}(A)\,\eta}{1 - \kappa_{\|\cdot\|}(A)\,\eta} \quad \text{whenever } \kappa_{\|\cdot\|}(A)\,\eta < 1.$$

Theorem 3 (Perturbation of inverses (Neumann series bound))**.** *If $\|A^{-1}E\| < 1$, then $(A + E)^{-1}$ exists and*

$$\|(A+E)^{-1}\| \le \frac{\|A^{-1}\|}{1 - \|A^{-1}E\|}, \qquad \|(A+E)^{-1} - A^{-1}\| \le \frac{\|A^{-1}\|^2\,\|E\|}{1 - \|A^{-1}E\|}.$$

D. Links to Singular Values & Eigenvalues

- $\kappa_2(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$ for any nonsingular A.
- **Normal/Symmetric:** If A is normal ($AA^* = A^*A$), then $\|A\|_2 = \max_i |\lambda_i(A)|$. For symmetric positive definite (SPD) A,

$$\kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}.$$

- **Weyl inequalities (singular values):** For any E, $|\sigma_i(A+E) - \sigma_i(A)| \le \|E\|_2$ for all i.
- **Hoffman–Wielandt:** $\sum_i \big(\sigma_i(A+E) - \sigma_i(A)\big)^2 \le \|E\|_F^2$.

E. "Affine vs. Linear" Transform Lemma (Scale Invariance; Offset Perturbation)

Lemma 1 (Scaling invariance)**.** *For any nonzero scalar α and any consistent matrix norm, $\kappa(\alpha A) = \kappa(A)$.*

Proof sketch. $\|\alpha A\| = |\alpha| \, \|A\|$ and $(\alpha A)^{-1} = \alpha^{-1} A^{-1}$ yield $\|\alpha A\| \, \|(\alpha A)^{-1}\| = \|A\| \, \|A^{-1}\|$. □

Lemma 2 (Affine shift as rank-1 perturbation)**.** *Let $C = \alpha A + \beta \, \boldsymbol{u}\boldsymbol{v}^T$ with $\alpha \neq 0$. Then $C = \alpha\big(A + E\big)$ with $E = (\beta/\alpha) \, \boldsymbol{u}\boldsymbol{v}^T$. Thus $\kappa_2(C) = \kappa_2(A + E)$ and, by Weyl,*

$$\frac{\sigma_{\max}(A) - \|E\|_2}{\sigma_{\min}(A) + \|E\|_2} \;\leq\; \kappa_2(A + E) \;\leq\; \frac{\sigma_{\max}(A) + \|E\|_2}{\sigma_{\min}(A) - \|E\|_2},$$

whenever $\sigma_{\min}(A) > \|E\|_2$. For $E = \gamma \, \mathbf{1}\mathbf{1}^T$, $\|E\|_2 = |\gamma| \sqrt{mn}$.

F. Special Matrix Families: Closed Forms/Bounds

Diagonal $D = \mathrm{diag}(d_1, \ldots, d_n)$ (nonsingular).

$$\kappa_2(D) = \frac{\max_i |d_i|}{\min_i |d_i|}, \qquad \kappa_1(D) = \kappa_\infty(D) = \kappa_2(D),$$

$$\kappa_F(D) = \Big(\sum_{i=1}^{n} d_i^2\Big)^{1/2} \Big(\sum_{i=1}^{n} d_i^{-2}\Big)^{1/2}.$$

Orthogonal/Unitary Q ($Q^T Q = I$).

$$\kappa_2(Q) = 1, \qquad \|Q\|_F = \sqrt{n}, \quad \kappa_F(Q) = \|Q\|_F \left\|Q^{-1}\right\|_F = n.$$

Symmetric Positive Definite (SPD).

$$\kappa_2(A) = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}, \qquad \|A\|_2 = \lambda_{\max}(A), \quad \left\|A^{-1}\right\|_2 = \frac{1}{\lambda_{\min}(A)}.$$

Triangular matrices T. Exact $\kappa_2(T)$ is generally nontrivial; practical bounds use $\|T\|_1$ or $\|T\|_\infty$:

$$\kappa_2(T) \;\le\; \sqrt{\kappa_1(T)\,\kappa_\infty(T)}, \qquad \kappa_p(T) = \|T\|_p \|T^{-1}\|_p.$$

Growth-factor considerations in Gaussian elimination (without pivoting) relate to $\|L\|, \|U\|$ bounds (see Higham, Ch. 9).

Normal matrices (incl. symmetric, orthogonal, unitary).

$$\|A\|_2 = \max_i |\lambda_i(A)|, \qquad \kappa_2(A) = \frac{\max_i |\lambda_i(A)|}{\min_i |\lambda_i(A)|} \quad \text{if } A \text{ is SPD.}$$

(Brief) Tridiagonal/Toeplitz observations. For well-conditioned tridiagonal SPD matrices (e.g., arising from 1D Poisson), κ_2 grows like $\mathcal{O}(n^2)$ in mesh-refinement analogs; structured solvers exploit bandedness. Specific closed forms depend on coefficients; see Golub–Van Loan, §4.5 & §4.9.

G. Frequently Used Inequalities & Identities (Quick Citables)

- **Submultiplicativity:** $\|AB\| \le \|A\|\,\|B\|$; hence $\kappa(AB) \le \kappa(A)\kappa(B)$ for induced norms.
- **Spectral/Frobenius:** $\|A\|_2 \le \|A\|_F \le \sqrt{\operatorname{rank}(A)}\;\|A\|_2$.
- **Singular-value sandwich:** $|\sigma_i(A+E) - \sigma_i(A)| \le \|E\|_2$ (Weyl).
- **Hoffman–Wielandt:** $\sum_i (\sigma_i(A+E) - \sigma_i(A))^2 \le \|E\|_F^2$.
- **Rayleigh quotient (SPD):** $\lambda_{\min}(A)\,\|x\|_2^2 \le x^T A x \le \lambda_{\max}(A)\,\|x\|_2^2$.
- **Resolvant/inverse perturbation:** If $\|A^{-1}E\| < 1$, then $(A+E)^{-1}$ exists and $\|(A+E)^{-1}\| \le \|A^{-1}\|/(1 - \|A^{-1}E\|)$.
- **Conditioning under scaling:** $\kappa(\alpha A) = \kappa(A)$ for any $\alpha \neq 0$ (all consistent norms).

H. Notes for Practice (Unit/Offset Invariance)

Unit changes: Pure rescalings leave κ unchanged; affine unit shifts (e.g., °F→°C) act as rank-1 perturbations and can change κ. For data matrices, *center* (and optionally standardize) before forming Gram/covariance matrices to make results unit-invariant.

References (Books)

- N. J. Higham, *Accuracy and Stability of Numerical Algorithms* (2nd ed.), SIAM, 2002.
- G. H. Golub and C. F. Van Loan, *Matrix Computations* (4th ed.), Johns Hopkins Univ. Press, 2013.
- L. N. Trefethen and D. Bau III, *Numerical Linear Algebra*, SIAM, 1997.